



MISO DPP 2017 February Central Area Study Phase III Final Report

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1. EXECUTIVE SUMMARY

This report presents the results of a System Impact Study (SIS) performed to re-evaluate the interconnection of the generators in the DPP 2017 February Central Area Phase III (Central Area DPP3). The study was performed under the direction of MISO and reviewed by an ad hoc study group. The ad hoc study group was formed to review the study scope, methodology, models and results. The ad hoc study group consisted of representatives from the interconnection customers and the following utility companies – Ameren, American Electric Power, Big Rivers Electric Corporation, City of Springfield (IL) Water Light & Power, Columbia (MO) Water and Light, Commonwealth Edison, Duke Energy Midwest, Hoosier Energy, Indianapolis Power & Light, MISO, Northern Indiana Public Service, PJM, Southern Illinois Power Cooperative, and Vectren.



1.1. PROJECT LIST

The original interconnection requests for DPP 2017 February Central Area had twelve (12) generation projects. Project J658 was withdrawn prior the beginning of Phase I study and was not considered in the results. Project J782 was withdrawn during Decision Point I and was not considered in the Phase III results. Additionally, J734, J759, and J783 reduced their requested NRIS levels during Decision Point I to 11.5 MW, 69.9 MW, and 69.9 MW respectively. Therefore, there are ten (10) generation projects with a combined nameplate rating of 2296.6 MW (ERIS) / 2307.9 MW (NRIS). The detailed list of projects in DPP 2017 Central Area is shown in Table 1-1, and the Phase III study was kicked off on September 19th, 2018.

Table 1-1 List of DPP February 2017 Central Area Phase III Projects

Project	Fuel Type	TO	County	State	Service Requested	MW	POI
J708	Combined Cycle	SIGE	Posey County	IN	NRIS	847	AB Brown 138 kV sub
J734	Gas	AMRN	Ford County	IL	NRIS Only	11.5	Gibson City South Substation
J740	Wind	NIPSCO	Jasper County, Pulaski County	IN	NRIS	200	Reynold 345kV substation
J753	Solar	BREC	Breckinridge County	KY	NRIS	100	Hardinsburg Sub 161kV
J754	Wind	DEI	Montgomery County	IN	NRIS	303.6	Cayuga - Nucor 345kV
J756	Wind	AMRN	Logan County	IL	NRIS	202.4	Mason City to Fogarty 138kV
J757	Wind	AMRN	Morgan County, Sangamon County	IL	NRIS	303.6	Austin-Meredosia 345kV line
J759	Solar	HE	Spencer County	IN	NRIS	70 ERIS, 69.9 NRIS	Troy 161kV Sub
J762	Solar	BREC	Meade County	KY	NRIS	200	Meade Sub 161kV
J783	Solar	SIGE	Spencer County	IN	NRIS	70 ERIS, 69.9 NRIS	Grandview Sub



1.2. TOTAL NETWORK UPGRADES

The cost allocation of Network Upgrades for the projects in the DPP 2017 February Central Cycle Phase III reflects responsibilities for mitigating system impacts. The total cost of network upgrades is listed in Table 1-2 as shown below. The costs for Network Upgrades are planning level estimates and subject to be revised in the facility studies.

Table 1-2 Total Cost of Network Upgrades for DPP 2017 February Central Phase III Projects

Project	ERIS Network Upgrades (\$)				NRIS Network Upgrades (\$)	Interconnection Facilities (\$)		Shared Network Upgrade (\$)	Total Network Upgrade Cost (\$)
	Steady State	Stability	Short Circuit	Affected System	Deliverability	TO Network Upgrades	TO - Owned Direct Assigned		
a	b	c	d	e	f	g	h	i	j = b+c+d+f+g+i
J708	\$1,600,000	\$0	\$7,286,034	\$5,936,000	\$1,330,998	\$1,040,259	\$0	\$0	\$11,257,291
J734	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
J740	\$0	\$0	\$0	\$182,706	\$0	\$3,865,876	\$0	\$0	\$3,865,876
J753	\$0	\$0	\$0	\$30,061	\$0	\$200,000	\$0	\$0	\$200,000
J754	\$0	\$0	\$0	\$ 9,642,760	\$0	\$14,832,311	\$1,308,575	\$0	\$14,832,311
J756	\$0	\$0	\$0	\$0	\$0	\$7,249,000	\$500,000	\$0	\$7,249,000
J757	\$0	\$0	\$0	\$248,536*	\$0	\$8,500,000#	\$500,000	\$0	\$8,500,000
J759	\$0	\$0	\$0	\$0	\$0	\$1,689,000	\$0	\$0	\$1,689,000
J762	\$0	\$0	\$0	\$44,939	\$0	\$500,000	\$0	\$0	\$500,000
J783	\$480,000	\$0	\$0	\$0	\$64,002	\$885,456	\$0	\$0	\$1,429,458

* PJM is investigating new constraint of Casey – Sullivan 345 kV and J757's impacts.

Awaiting draft Interconnection Customer Interconnection Facility Facilities Study (IC IF FS)

2. ERIS ANALYSIS

2.1. J708, J740, J753, J754, J759, J762 AND J783 SYSTEM IMPACT STUDY

J708, J740, J753, J754, J759, J762 and J783 system impact study were performed by ABB under the direction of MISO and ad hoc group to identify network upgrades required to interconnect those requests in the DPP 2017 February Central Area group to the transmission system. System constraints were identified and the planning level cost estimates for the network upgrade are shown in Table 2-1. The detailed study report is listed in Appendix A.



Table 2-1 Planning Level Cost Estimation

Project	ERIS Network Upgrades (\$)			Interconnection Facilities (\$)		Total Network Upgrade Cost(\$)
	Steady State	Stability	Short Circuit	TO Network Upgrades	TO - Owned Direct Assigned	
J708	\$1,600,000	\$0	\$11,500,000	\$8,326,292	\$0	\$22,757,290
J740	\$0	\$0	\$0	\$3,865,876	\$0	\$3,865,876
J753	\$0	\$0	\$0	\$200,000	\$0	\$200,000
J754	\$0	\$0	\$0	\$19,654,620	\$0	\$19,654,620
J759	\$0	\$0	\$0	\$1,689,000	\$0	\$1,689,000
J762	\$0	\$0	\$0	\$500,000	\$0	\$500,000
J783	\$480,000	\$0	\$0	\$885,456	\$0	\$1,365,456

2.2. J756 and J757 SYSTEM IMPACT STUDY

J756 and J757 system impact study was performed by Ameren under the direction of MISO and ad hoc group to identify network upgrades required to interconnect those requests in the DPP 2017 February Central Area group to the transmission system. The required system upgrades are identified in the analysis and the planning level cost estimates are shown in Table 2-2. The detailed study report is listed in Appendix B.

Table 2-2 Planning Level Cost Estimation

Project	ERIS Network Upgrades (\$)			Interconnection Facilities (\$)		Total Network Upgrade Cost (\$)
	Steady State	Stability	Short Circuit	TO Network Upgrades	TO - Owned Direct Assigned	
J756	\$0	\$0	\$0	\$7,249,000	\$500,000	\$7,249,000
J757	\$0	\$0	\$0	\$8,500,000	\$500,000	\$8,500,000

2.3. AFFECTED SYSTEM IMPACT STUDY

J708

The PJM study identified that project J708 contributes to the overload of the following projects:

1. Sullivan – Casey 345 kV

Per PJM cost allocation rules, project J708 receives cost allocation for upgrades required to mitigate the above listed constraints. The planned upgrade is a new 765 kV line from Sullivan – Reynolds (PJM Network Upgrade N5034). Cost estimate is \$464 million. The project J708 is allocated 1.3% of the cost for \$5.936 million.

J734

No mitigations were found to be required.



J740

The PJM study identified that project J740 contributes to the overload of the following project:

1. Twin Branch – Argenta 345 kV

Per PJM cost allocation rules, project J740 receives cost allocation for upgrades required to mitigate the above listed constraint. The planned upgrade is to replace a wavetrapp at Twin Branch (PJM Upgrade N5310.1), perform a sag check (N5240), and replace a substation structure at Argenta (N5310.2). Cost estimate is \$900,000. The project J740 is allocated 20.3% of the cost for \$182,706.

J753

The ABB study identified that project J753 contributes to the overload of the following project:

1. New Hardinsburg – Hardinsburg 138 kV

Project J753 receives cost allocation for upgrades required to mitigate the above listed constraint. Cost estimate is \$75,000. The project J753 is allocated 40% of the cost for \$30,061.

J754

The PJM study identified that project J754 contributes to the overload of the following projects:

1. Dequin – Meadow 345 kV Circuit 1
2. Dequin – Meadow 345 kV Circuit 2
3. Sullivan – Casey 345 kV Circuit 1

Per PJM cost allocation rules, project J754 receives cost allocation for upgrades required to mitigate the above listed constraints. The planned upgrade is a new 765 kV line from Sullivan – Reynolds (PJM Network Upgrade N5034). Cost estimate is \$464 million. The project J754 is allocated 2.0% of the cost for \$9.427 million.

The PJM study identified that project J754 contributes to the overload of the following project:

2. Eugene – DeQuin 345 kV

The planned upgrade is to reconductor the line (PJM Upgrade B2777). The in-service date of the project is 2021. An interim study will need to be performed if project J754 comes into service prior to PJM Upgrade B2777.

The PJM study identified that project J754 contributes to the overload of the following project:

3. Twin Branch – Argenta 345 kV

Per PJM cost allocation rules, project J754 receives cost allocation for upgrades required to mitigate the above listed constraint. The planned upgrade is to replace a wavetrapp at Twin Branch (PJM Upgrade N5310.1), perform a sag check (N5240), and replace a substation structure at Argenta (N5310.2). Cost estimate is \$900,000. The project J754 is allocated 23.97% of the cost for \$215,760.



J756

No mitigations were found to be required.

J757

The PJM study identified that project J757 contributes to the overload of the following project:

1. Casey – Sullivan 345 kV

Per PJM cost allocation rules, project J757 receives cost allocation for upgrades required to mitigate the above listed constraint. The proposed upgrade is to rebuild or re-conductor the 345 kV line. Cost estimate is \$20-30 million. The project J757 allocation is yet to be determined.

The PJM study identified that project J757 contributes to the overload of the following project:

1. Twin Branch – Argenta 345 kV

Per PJM cost allocation rules, project J757 receives cost allocation for upgrades required to mitigate the above listed constraint. The planned upgrade is to replace a wavetrap at Twin Branch (PJM Upgrade N5310.1), perform a sag check (N5240), and replace a substation structure at Argenta (N5310.2). Cost estimate is \$900,000. The project J757 is allocated 27.62% of the cost for \$248,536.

J759

No mitigations were found to be required.

J762

The ABB study identified that project J762 contributes to the overload of the following project:

1. New Hardinsburg – Hardinsburg 138 kV

Project J762 receives cost allocation for upgrades required to mitigate the above listed constraint. Cost estimate is \$75,000. The project J762 is allocated 60% of the cost for \$44,939.

J783

No mitigations were found to be required.



3. DELIVERABILITY ANALYSIS

3.1. INTRODUCTION

Generator interconnection requests have to pass Generator Deliverability Study to be granted Network Resource Interconnection Services (NRIS). If the generator is determined as not fully deliverable, the customer can either choose to elect the amount of NRIS available without upgrades or build system upgrades that will make the generator fully deliverable. Generator Deliverability Study ensures that the Network Resources, on an aggregate basis, can meet the MISO aggregate load requirements during system peak condition without getting bottled up.

MISO Generator Deliverability Study whitepaper describing the algorithm can be found at http://www.midwestmarket.org/publish/Document/3e2d0_106c60936d4_-767f0a48324a

3.2. DETERMINING THE MW RESTRICTION

If one facility is overloaded based on the assessed “severe yet credible dispatch” scenario described in the study methodology, and the generator under study is in the “Top 30 DF List” (see white paper for detail), part or all of its output is not deliverable. The restricted MW is calculated as following:

$$(\text{MW restricted}) = (\text{worst loading} - \text{MW rating}) / (\text{generator sensitivity factor})$$

If the result is larger than the maximum output of the generator, 100% of this generator’s output is not deliverable.

The generator is also responsible for any NEW base case (pre-shift) overload or NEW “severe yet credible dispatch overload” where the generator is not in the “Top 30 DF List”, if the generator’s DF is greater than 5%. Please see white paper for detail. The formula above also applies to these situations.



3.3. DELIVERABILITY STUDY RESULT

3.3.1. Deliverability Results Based on 2022 Summer Peak Case

J708

J708 is found 90.44% (766.0 MW) deliverable. Detailed study results are included in Table 3-1.

Table 3-1 J708 Deliverability Summary

J708 Deliverable (NRIS) Amount in 2022 Case: (Conditional on ERIIS upgrades and case assumptions)		766.0 MW (90.44%)					
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIIS Analysis?	Projects Associated With ERIS Constraint	Projects Associated With NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade
AEP Tap – Grand 138 kV	766.0	12.78%			J708	\$80,000	\$80,000
Northeast – Hillside 69 kV	779.5	6.14%			J708	\$1,015,000	\$1,015,000
Grand – Newtonville 138 kV	812.9	12.08%			J708, J783	\$235,998	\$300,000

J734

J734 is found 100% (11.5 MW) deliverable. Detailed study results are included in Table 3-2.

Table 3-2 J734 Deliverability Summary

J734 Deliverable (NRIS) Amount in 2022 Case: (Conditional on ERIIS upgrades and case assumptions)	11.5 MW (100%)
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J740

J740 is found 100% (200MW) deliverable. Detailed study results are included in Table 3-3.

Table 3-3 J740 Deliverability Summary

J740 Deliverable (NRIS) Amount in 2022 Case: (Conditional on ERIIS upgrades and case assumptions)	200 MW (100%)
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J753

J753 is found 100% (100 MW) deliverable. Detailed study results are included in Table 3-4.

Table 3-4 J753 Deliverability Summary

J753 Deliverable (NRIS) Amount in 2022 Case: (Conditional on ERIIS upgrades and case assumptions)	100 MW (100%)
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J754

J754 is found 100% (303.6 MW) deliverable. Detailed study results are included in Table 3-5.

Table 3-5 J754 Deliverability Summary

J754 Deliverable (NRIS) Amount in 2022 Case: (Conditional on ERIIS upgrades and case assumptions)	303.6 MW (100%)
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J756

J756 is found 100% (202.4 MW) deliverable. Detailed study results are included in Table 3-6.

Table 3-6 J756 Deliverability Summary

J756 Deliverable (NRIS) Amount in 2022 Case: (Conditional on ERIIS upgrades and case assumptions)	202.4 MW (100%)
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J757

J757 is found 100% (303.6 MW) deliverable. Detailed study results are included in Table 3-7.

Table 3-7 J757 Deliverability Summary

J757 Deliverable (NRIS) Amount in 2022 Case: (Conditional on ERIIS upgrades and case assumptions)	303.6 MW (100%)
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J759

J759 is found 100% (69.9 MW) deliverable. Detailed study results are included in Table 3-8.

Table 3-8 J759 Deliverability Summary

J759 Deliverable (NRIS) Amount in 2022 Case: (Conditional on ERIIS upgrades and case assumptions)	69.9 MW (100%)
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J762

J762 is found 100% (200 MW) deliverable. Detailed study results are included in Table 3-9.

Table 3-9 J762 Deliverability Summary

J762 Deliverable (NRIS) Amount in 2022 Case: (Conditional on ERIIS upgrades and case assumptions)	200 MW (100%)
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J783

J783 is found 96.09% (67.2 MW) deliverable. Detailed study results are included in Table 3-11.

Table 3-11 J783 Deliverability Summary

J783 Deliverable (NRIS) Amount in 2022 Case: (Conditional on ERIIS upgrades and case assumptions)	67.2 MW (93.56%)						
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIIS Analysis?	Projects Associated With ERIS Constraint	Projects Associated With NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade
Grand – Newtonville 138 kV	67.2	39.65%			J708, J783	\$64,002	\$300,000



3.3.2. Deliverability Network Upgrades Summary

The cost estimate for proposed network upgrades are listed in Table 3-12.

Table 3-12 NRIS Constraints and Cost Estimate

Monitored Element	Mitigation	Planning Level Cost	Queue Projects with Impacts
Northeast – Hillside 69 kV	Rebuild 1.5 miles of 69 kV	\$1,015,000	J708
AEP Tap – Grand 138 kV	Upgrade terminal equipment	\$80,000	J708
Grand – Newtonville 138 kV	Replace breaker and switches	\$300,000	J708, J783

4. SHARED NETWORK UPGRADES ANALYSIS

Shared Network Upgrade (SNU) test for Network Upgrades driven by higher queued interconnection project was performed for this System Impact Study. No SNU were found for DPP February 2017 Central Area Projects.

5. COST ALLOCATION

The cost allocation of Network Upgrades for the study group reflects responsibilities for mitigating system impacts based on Interconnection Customer-elected level of Network Resource Interconnection service as of the draft System Impact Study report date.

5.1. COST ASSUMPTIONS FOR NETWORK UPGRADES

The cost estimate for each network upgrade identified in System Impact Study was provided by the corresponding transmission owning company.



5.2. COST ALLOCATION METHODOLOGY

The costs of Network Upgrades (NU) for a set of generation projects (one or more sub-groups or entire group with identified NU) are allocated based on the MW impact from each project on the constrained facilities in the Post Case.

Cost Allocation Methodology for Thermal Constraint

1. With all Group Study generation projects dispatched in the Post Case, all thermal constraints are identified.
2. Distribution factor from each project on each constraint is obtained.
3. For each thermal constraint, the maximum MW contribution (increasing flow) from each project is then calculated in the Post Case without any network upgrades.
4. For each thermal constraint, the cost estimates for one or a subset of NU are provided by the corresponding transmission owning company.
5. Then the cost of each NU is allocated based on the pro rata share of the MW contribution from each project on the constraints mitigated or partly mitigated by this NU. The methodology to determine the cost allocation of one NU is:

Project A cost portion of NU

$$= \text{Cost of NU} * \frac{\text{Max(Proj. A MW contribution on constraint)}}{\sum_i \text{Max(Proj. i MW contribution on constraint)}}$$

6. The total NU costs for each project are calculated if more than one NU are required.



APPENDICES

APPENDIX A – APPENDIX A - J708, J740, J753, J754, J759, J762, AND J783 ERS ANALYSIS STUDY REPORT

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APPENDIX F – DPP FEBRUARY 2017 DELIVERABILITY DETAILED RESULTS